
Reasoning Trace Length as a Proxy for Robustness: A Controlled Study of Fixed and Adaptive Inference Policies

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Abstract

Chain-of-thought reasoning is widely assumed to improve large language model performance, yet the relationship between reasoning trace length and out-of-distribution (OOD) robustness remains poorly understood. We present a controlled empirical study that systematically varies reasoning trace length via five prompt-based policies—NONE, SHORT, MEDIUM, LONG, and an uncertainty-triggered ADAPTIVE policy—while holding the base model, temperature, and evaluation pipeline constant. Across 1,250 evaluations on five benchmarks spanning an in-distribution to OOD gradient, we find a strong non-monotonic relationship: medium-length traces yield the best in-distribution accuracy (90%), while the LONG policy degrades both in-distribution (80%) and OOD performance (40% on MATH-500 vs. 46% for SHORT). The ADAPTIVE policy achieves the best OOD-near accuracy (56%) and the smallest robustness gap (30 percentage points), but is limited by poor confidence calibration ($ECE = 0.258$). Token-normalized efficiency favors concise reasoning by up to $70\times$ over verbose reasoning. A confirmatory experiment on GPT-4.1-MINI amplifies these effects: OOD-near accuracy collapses from 70% (SHORT) to 15% (LONG). Our results challenge the “more reasoning is better” assumption and provide actionable guidance for inference-time compute allocation.

1 Introduction

Should language models think longer to answer harder questions? The prevailing wisdom in chain-of-thought (CoT) reasoning research suggests yes: longer, more detailed reasoning traces tend to improve accuracy on challenging benchmarks [Wei et al., 2022, Wang et al., 2022]. This assumption underpins the design of recent reasoning-focused models and inference-time scaling strategies [Aggarwal and Welleck, 2025]. Yet a growing body of evidence suggests the relationship between reasoning depth and performance is more nuanced—particularly when models encounter inputs that differ from their training distribution.

Why does trace length matter for robustness? As LLMs are deployed in high-stakes reasoning tasks spanning medical diagnosis, legal analysis, and scientific discovery, their reliability under distribution shift becomes critical. Practitioners must decide how much inference-time compute to allocate: too little and the model may fail on complex problems; too much and the model may “over-think,” accumulating errors and hallucinations that degrade performance [Su et al., 2025, Shojaei et al., 2025]. Recent work has shown that longer CoT traces can hurt accuracy on both easy [Wu et al., 2025] and out-of-distribution tasks [Li et al., 2025], but no study has systematically isolated the causal effect of trace length on OOD robustness under controlled conditions.

What is missing? Prior work has studied the length–accuracy tradeoff [Wu et al., 2025, Jin et al., 2024] and OOD fragility of CoT [Li et al., 2025, Wang et al., 2024] as separate phenomena. Length-controlled studies typically evaluate only in-distribution performance [Aggarwal and Welleck, 2025, Lee et al., 2025], while robustness studies rarely control for trace length. The interaction between

these two dimensions—how trace length modulates robustness—remains an open question. Furthermore, adaptive trace-length policies conditioned on model uncertainty have been proposed conceptually [Zhang et al., 2025] but not compared against strong fixed-length baselines under matched evaluation conditions.

Our approach. We design a controlled experiment that isolates reasoning trace length as the sole independent variable. We define five prompt-based policies—NONE (direct answer), SHORT (1–2 steps), MEDIUM (4–6 steps), LONG (8+ steps), and ADAPTIVE (short with confidence-triggered retry)—and evaluate each on 250 questions spanning five benchmarks from in-distribution arithmetic (GSM8K) to far-ODD commonsense reasoning (COMMONSENSEQA). All experiments use deterministic decoding (temperature = 0) on GPT-4.1-NANO, with a confirmatory replication on GPT-4.1-MINI.

Our key findings are striking. In-distribution accuracy follows an inverted U-shape, peaking at SHORT/MEDIUM (90%) and declining at LONG (80%). OOD performance degrades monotonically with trace length: SHORT achieves 46% on MATH-500 while LONG achieves only 40%, a gap that widens to 70% vs. 15% on the stronger GPT-4.1-MINI model. The ADAPTIVE policy achieves the best OOD-near accuracy (56%) and the smallest robustness gap (30 percentage points), while a $70\times$ efficiency advantage of concise over verbose reasoning highlights the cost of overthinking.

We make the following contributions:

- We conduct the first controlled study of reasoning trace length as a causal factor in OOD robustness, demonstrating a strong non-monotonic relationship across five benchmarks and two model sizes.
- We show that verbose reasoning (LONG) catastrophically degrades OOD performance—by up to 55 percentage points on GPT-4.1-MINI—providing empirical support for the “overthinking” hypothesis.
- We evaluate an uncertainty-adaptive trace-length policy that achieves the best OOD-near accuracy and smallest robustness gap, while identifying confidence calibration as its key bottleneck.
- We quantify the cost–accuracy–robustness tradeoff, showing that short reasoning dominates long reasoning in token-normalized efficiency by up to $70\times$.

Paper organization. Section 2 positions this work against prior research. Section 3 details the experimental protocol. Section 4 reports quantitative findings. Section 5 discusses implications and limitations. Section 6 concludes.

2 Related Work

Chain-of-thought reasoning and trace length. Chain-of-thought prompting [Wei et al., 2022] demonstrated that eliciting intermediate reasoning steps substantially improves LLM performance on multi-step tasks. Subsequent work explored how the *length* of these traces affects accuracy. Jin et al. [2024] showed that longer reasoning steps generally improve performance, even when the additional content is not strictly informative. Wu et al. [2025] formalized this relationship, proving an inverted U-shaped accuracy curve with respect to trace length and deriving a closed-form expression for the optimal length as a function of task difficulty and model capability. Su et al. [2025] complemented this with an empirical characterization of “overthinking” and “underthinking,” finding a strong negative correlation ($r = -0.72$) between response length and accuracy on incorrect answers. Our work extends these findings by studying how the length–accuracy relationship changes under distribution shift—a dimension these prior studies did not examine.

Overthinking and efficient reasoning. The observation that longer reasoning can hurt performance has motivated a growing literature on efficient inference. Shojaei et al. [2025] identified three performance regimes—low, medium, and high complexity—where reasoning overhead is counterproductive, beneficial, and ineffective, respectively. Aggarwal and Welleck [2025] proposed L1/LCPO, an RL-based method for training models with controllable reasoning length, discovering that adaptive length allocation consistently outperforms fixed-length budgets. Lee et al. [2025] introduced “token complexity” as the minimal tokens required per problem, revealing sharp accuracy thresholds. Compression approaches such as C3oT [Kang et al., 2024] and TokenSkip [Xia et al., 2025] achieve 40–50% length reduction with minimal accuracy loss. Unlike these works, which focus on in-distribution efficiency, we study how length control affects robustness to distribution shift.

Robustness of reasoning traces. Several studies have examined the fragility of CoT reasoning under perturbation. Wang et al. [2024] showed that noisy rationales—both irrelevant and inaccurate—degrade CoT performance, with longer chains being more susceptible to error accumulation. von Recum et al. [2026] found that reasoning LLMs are generally robust to perturbations but that adversarial noise inflates CoT length by over 200%. Li et al. [2025] argued that CoT effectiveness is governed by the distribution discrepancy between training and test data. Li et al. [2026] demonstrated that extended reasoning traces induce overconfidence, causing confidence-based defenses to fail. Our work bridges the trace-length and robustness literatures by directly measuring how fixed and adaptive length policies modulate OOD accuracy and calibration.

Adaptive inference-time compute. The idea of dynamically allocating reasoning budgets has gained traction. Zhang et al. [2025] surveyed adaptive approaches, distinguishing fixed-budget from dynamic-scaling strategies. Chen et al. [2025] showed that long CoT enables self-correction capabilities that transfer to OOD settings. Ballon et al. [2026] developed a protocol for probing reasoning traces at fixed token percentiles, finding that accuracy monotonically increases with trace percentage. Our ADAPTIVE policy differs from prior RL-based approaches [Aggarwal and Welleck, 2025] by using a simpler prompt-based mechanism with self-reported confidence, allowing us to study uncertainty-adaptive control without model retraining.

3 Methodology

3.1 Problem Formulation

We study the function $f : \mathcal{P} \times \mathcal{D} \rightarrow [0, 1]$ mapping a reasoning policy $p \in \mathcal{P}$ and a dataset $d \in \mathcal{D}$ to an accuracy score, where $\mathcal{P} = \{\text{NONE}, \text{SHORT}, \text{MEDIUM}, \text{LONG}, \text{ADAPTIVE}\}$ and $\mathcal{D}$ spans an in-distribution (ID) to out-of-distribution (OOD) gradient. We define the **robustness gap** as

$$\Delta(p) = f(p, d_{\text{ID}}) - f(p, d_{\text{OOD}}), \quad (1)$$

where lower values indicate better robustness. We also measure **token efficiency** $\eta(p) = \text{correct}(p)/\text{tokens}(p)$, the number of correct answers per 1,000 completion tokens.

3.2 Reasoning Policies

We define five prompt-based policies that control reasoning trace length while keeping the underlying model identical:

1. **NONE** (max 100 tokens): The model outputs only a direct final answer with no reasoning.
2. **SHORT** (max 300 tokens): At most 1–2 brief reasoning sentences before the answer.
3. **MEDIUM** (max 800 tokens): Standard chain-of-thought with 4–6 reasoning steps.
4. **LONG** (max 2,000 tokens): Detailed reasoning with 8+ steps including self-verification and error checking.
5. **ADAPTIVE**: First attempts SHORT reasoning with a self-reported confidence score. If confidence < 0.70 , retries with LONG reasoning. This tests whether uncertainty-aware escalation can combine the efficiency of short reasoning with the depth of long reasoning.

All policies share an identical answer extraction format (ANSWER: <response>), ensuring that accuracy differences reflect reasoning quality rather than parsing artifacts. We use deterministic decoding (temperature = 0) to eliminate sampling variance.

3.3 Datasets

We select five established benchmarks spanning an ID-to-OOD gradient (table 1). GSM8K serves as the in-distribution anchor: it requires multi-step arithmetic reasoning that aligns well with standard CoT prompting. MATH-500 represents near-OOD transfer (harder math, different problem styles). The three far-OOD datasets (ARC-CHALLENGE, MMLU-STEM, COMMONSENSEQA) test generalization to entirely different reasoning domains—science, STEM knowledge, and commonsense.

Table 1: Dataset characteristics. We sample 50 questions per dataset (random seed 42), totaling 250 unique questions. The ID/OOD classification reflects domain similarity to GSM8K.

Dataset	Domain	Format	n	Role
GSM8K	Grade-school math	Free-response	50	ID
MATH-500	Competition math	Free-response	50	OOD-near
ARC-CHALLENGE	Science QA	Multiple-choice	50	OOD-far
MMLU-STEM	Multi-subject STEM	Multiple-choice	50	OOD-far
COMMONSENSEQA	Commonsense reasoning	Multiple-choice	50	OOD-far

Table 2: Accuracy (%) by policy and dataset on GPT-4.1-NANO ($n = 50$ per cell). Best result per column in **bold**. GSM8K is in-distribution; MATH-500 is OOD-near; remaining datasets are OOD-far. The LONG policy underperforms SHORT on four of five benchmarks.

Policy	GSM8K (ID)	MATH-500 (OOD-n)	ARC-C (OOD-f)	MMLU-STEM (OOD-f)	CSQA (OOD-f)
NONE	28.0	22.0	74.0	36.0	84.0
SHORT	90.0	46.0	82.0	30.0	80.0
MEDIUM	90.0	52.0	72.0	36.0	66.0
LONG	80.0	40.0	72.0	32.0	58.0
ADAPTIVE	86.0	56.0	80.0	32.0	68.0

3.4 Evaluation Protocol

Answer extraction. We parse model outputs using regex-based extraction targeting the ANSWER: pattern, with fallback to the last line of the response. Answers are normalized by stripping whitespace, removing formatting artifacts, and applying numeric comparison for mathematical answers.

Metrics. We report: (1) **Accuracy**: exact-match after normalization; (2) **Robustness gap**: $\Delta = \text{Acc}_{\text{ID}} - \text{Acc}_{\text{OOD}}$; (3) **Token efficiency**: correct answers per 1,000 completion tokens; (4) **Calibration**: expected calibration error (ECE) and Brier score for the ADAPTIVE policy’s self-reported confidence.

Statistical testing. We use pairwise McNemar tests on shared items with Benjamini-Hochberg correction ($\alpha = 0.05$) for all policy comparisons.

3.5 Models and Implementation

Our primary experiment uses GPT-4.1-NANO (1,250 API calls: 5 policies $\times$ 250 questions). We run a confirmatory experiment on GPT-4.1-MINI (400 calls: 4 policies $\times$ 100 questions, 20 per dataset) to test whether findings generalize across model sizes. Both models use temperature = 0 for deterministic outputs. The evaluation harness is implemented in Python 3.12 with the OpenAI SDK, NumPy, SciPy, Matplotlib, and Seaborn. All outputs are logged to JSONL files with resume-capable deduplication.

4 Results

4.1 Main Results: Non-Monotonic Length–Accuracy Relationship

Table 2 presents accuracy by policy and dataset. In-distribution (GSM8K) accuracy follows an inverted U-shape: NONE (28%) $\rightarrow$ SHORT (90%) $\rightarrow$ MEDIUM (90%) $\rightarrow$ LONG (80%). The 10-point drop from MEDIUM to LONG confirms that excessive reasoning hurts even in-distribution performance.

On OOD-near data (MATH-500), accuracy peaks with ADAPTIVE (56%) and drops to 40% for LONG—a reversal from the assumption that more reasoning improves harder tasks. Among fixed policies, MEDIUM (52%) outperforms LONG (40%) by 12 percentage points. On OOD-far datasets, the pattern is consistent: LONG underperforms SHORT on ARC-CHALLENGE (72% vs. 82%), COMMONSENSEQA (58% vs. 80%), and MMLU-STEM (32% vs. 30%, within noise).

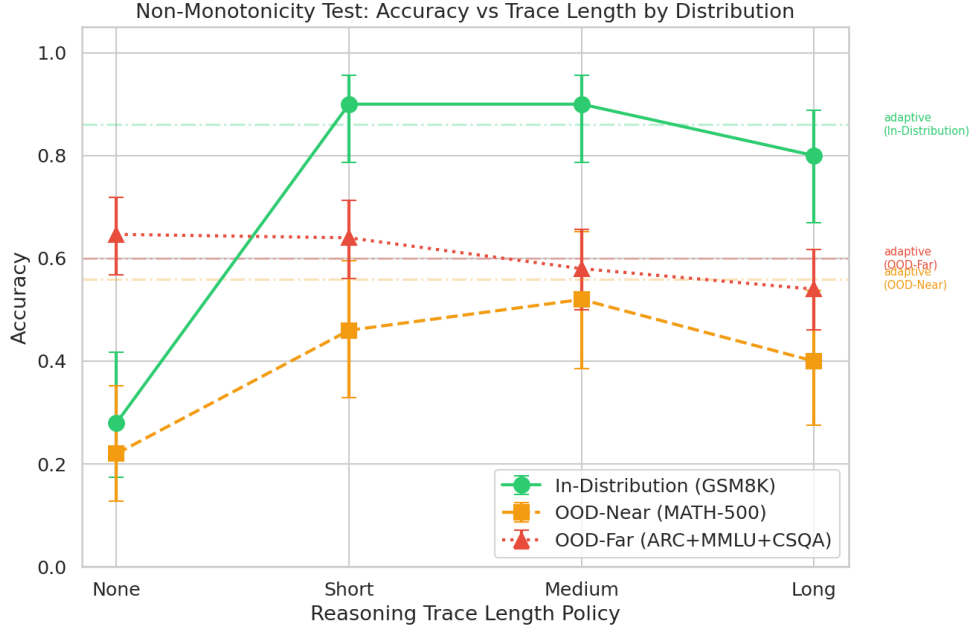


Figure 1: Accuracy vs. reasoning trace length by distribution type. ID accuracy follows an inverted U-shape, peaking at SHORT/MEDIUM and declining at LONG. OOD-near accuracy peaks at ADAPTIVE and drops monotonically for fixed policies beyond MEDIUM. OOD-far accuracy is highest for minimal or short reasoning.

Table 3: Robustness gap (ID accuracy – OOD accuracy, in percentage points). Lower is better. The ADAPTIVE policy achieves the smallest OOD-near gap. The negative gap for NONE on OOD-far reflects high baseline accuracy on multiple-choice tasks without reasoning.

Policy	ID Acc (%)	OOD-near Acc (%)	Gap (near)	Gap (far)
NONE	28.0	22.0	6.0	−36.7*
SHORT	90.0	46.0	44.0	26.0
MEDIUM	90.0	52.0	38.0	32.0
LONG	80.0	40.0	40.0	26.0
ADAPTIVE	86.0	56.0	30.0	26.0

*NONE’s negative far-OOD

gap arises because multiple-choice tasks have high baseline accuracy even without reasoning.

A notable finding is that NONE achieves 84% on COMMONSENSEQA—the highest score for any policy on that dataset. This suggests that for pattern-recognition tasks, reasoning introduces more noise than signal.

Figure 1 visualizes the non-monotonic relationship. The key observation is that the accuracy curve shifts leftward under distribution shift: optimal trace length is shorter for OOD data than for ID data.

4.2 Robustness Gap Analysis

Table 3 reports the robustness gap for each policy. The ADAPTIVE policy achieves the smallest OOD-near gap (30 percentage points), while SHORT has the largest (44 points). Among fixed policies, the gap does not monotonically decrease with length—MEDIUM (38 points) is better than SHORT (44 points), but LONG (40 points) is worse again.

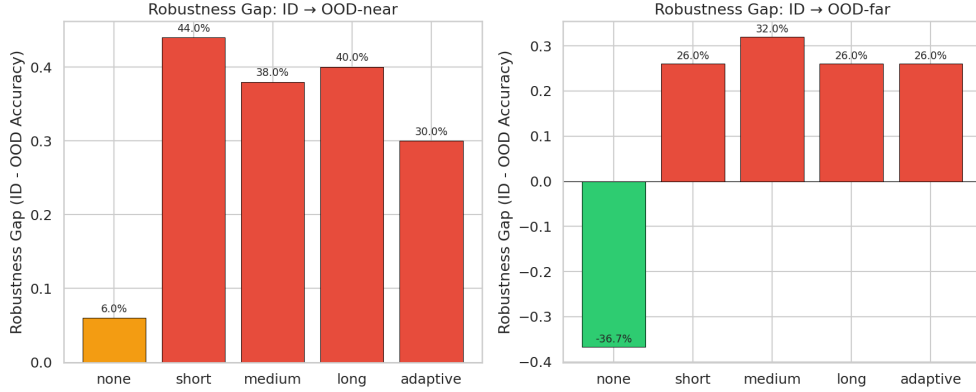


Figure 2: Robustness gap by policy. The ADAPTIVE policy achieves the smallest gap on OOD-near data, combining high ID accuracy with the best OOD-near transfer.

Table 4: Token usage and efficiency by policy. Efficiency is measured as correct answers per 1,000 completion tokens. SHORT achieves 10× the efficiency of LONG with higher overall accuracy.

Policy	Mean tokens	Median tokens	Efficiency (corr/1k tok)
NONE	9.4	6	52.11
SHORT	85.3	54	7.69
MEDIUM	315.1	228	2.01
LONG	765.5	647	0.74
ADAPTIVE	185.0	82	3.48

4.3 Token Efficiency

Table 4 shows that token-normalized efficiency strongly favors concise reasoning. The NONE policy achieves 52.1 correct answers per 1,000 tokens—70× more efficient than LONG (0.74). Even accounting for accuracy differences, SHORT (7.69) dominates LONG in cost-effectiveness.

The ADAPTIVE policy occupies an attractive position on the efficiency frontier (figure 3): it uses 185 tokens on average (vs. 766 for LONG) while achieving the best OOD-near accuracy. Only 4.4% of items triggered the long-retry path, suggesting the confidence threshold (0.70) was too conservative given the model’s overconfidence.

4.4 Statistical Significance

Pairwise McNemar tests with Benjamini-Hochberg correction yield the following significant comparisons at $\alpha = 0.05$:

- NONE vs. SHORT: $p < 0.0001$ (SHORT dramatically better)
- NONE vs. MEDIUM: $p = 0.0007$
- NONE vs. ADAPTIVE: $p < 0.0001$
- SHORT vs. LONG: $p = 0.020$ (SHORT significantly better than LONG)
- MEDIUM vs. LONG: $p = 0.062$ (trend, not significant after correction)

The SHORT vs. LONG comparison is particularly notable: short reasoning *significantly outperforms* long reasoning ($p = 0.020$), directly contradicting the “more reasoning is better” assumption.

4.5 Calibration Analysis

The ADAPTIVE policy relies on self-reported confidence to decide when to escalate. We find that this confidence is poorly calibrated: mean confidence is 0.924 while accuracy is only 0.644, yielding $ECE = 0.258$ and Brier score = 0.291. Figure 4 shows the reliability diagram. The model is

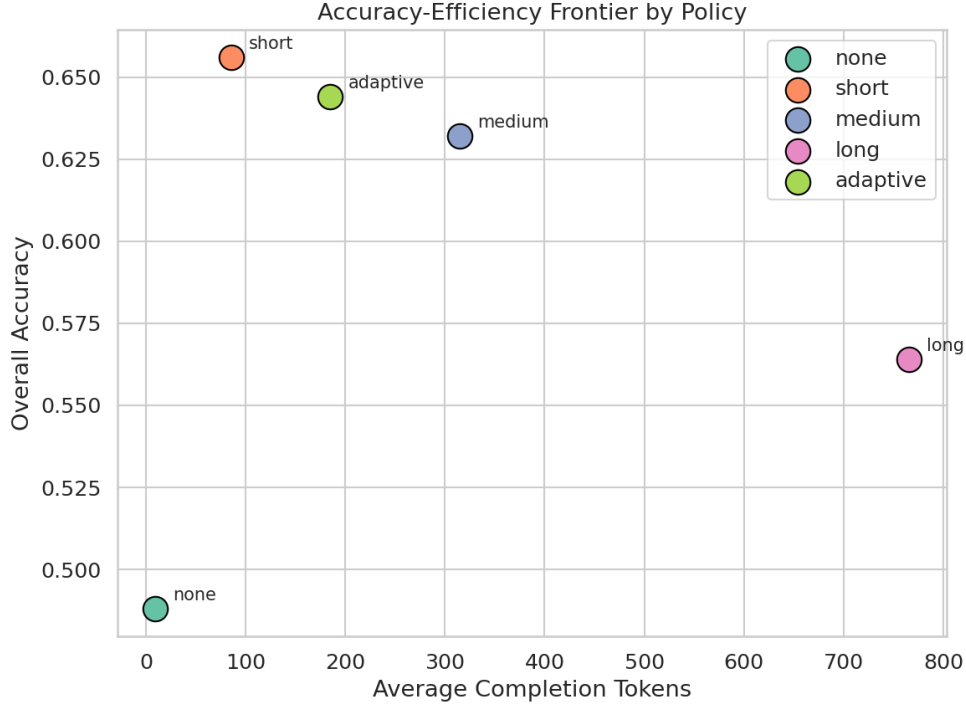


Figure 3: Accuracy–efficiency frontier. Each point represents a policy. SHORT and ADAPTIVE dominate LONG, achieving higher accuracy at a fraction of the token cost.

Table 5: Confirmatory results on GPT-4.1-MINI ($n = 20$ per dataset). The non-monotonic pattern is amplified: LONG collapses to 15% on OOD-near while SHORT achieves 70%. Best per column in **bold**.

Policy	Overall	ID	OOD-near	OOD-far
NONE	65.0	65.0	40.0	73.3
SHORT	81.0	95.0	70.0	80.0
MEDIUM	70.0	90.0	60.0	66.7
LONG	55.0	90.0	15.0	56.7

systematically overconfident across all confidence bins, which explains why only 4.4% of items triggered long-retry—the model rarely reports uncertainty even when it should.

4.6 Confirmatory Experiment

We replicate the core experiment on GPT-4.1-MINI with 4 policies and 20 questions per dataset (table 5). The non-monotonic pattern is dramatically amplified: OOD-near accuracy collapses from 70% (SHORT) to 15% (LONG)—a 55 percentage point gap. ID accuracy remains high (90–95%) for all CoT policies, indicating that the damage from verbose reasoning is specific to OOD transfer. This suggests that stronger models may be *more* susceptible to overthinking artifacts.

5 Discussion

Interpretation of non-monotonicity. Our results demonstrate that the relationship between reasoning trace length and accuracy is fundamentally different in-distribution versus out-of-distribution. In-distribution, moderate reasoning (SHORT, MEDIUM) provides the structure needed for multi-step problem solving. Out-of-distribution, additional reasoning steps appear to amplify domain-specific

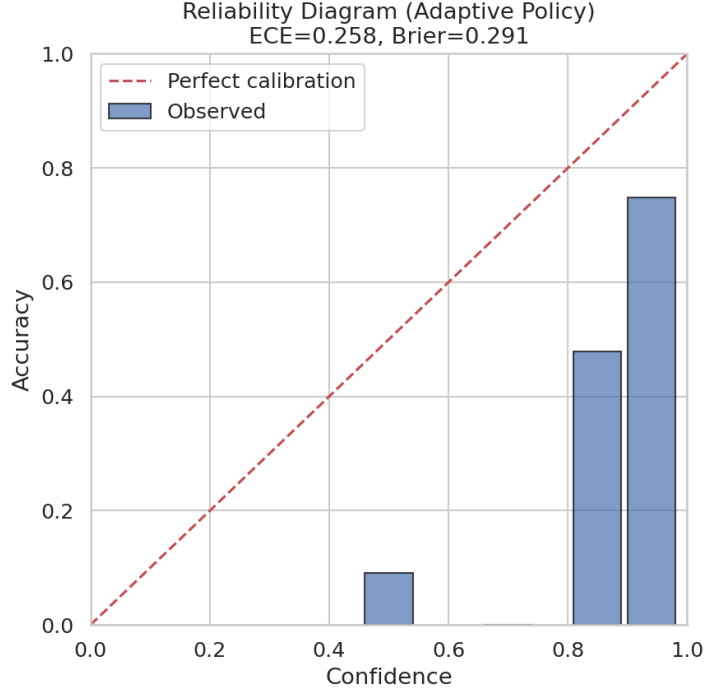


Figure 4: Reliability diagram for the ADAPTIVE policy. The model is systematically overconfident: mean self-reported confidence (0.924) far exceeds accuracy (0.644). Better calibration would improve adaptive escalation decisions.

errors and hallucinations, causing the model to “talk itself out” of correct initial intuitions. This is most dramatic on COMMONSENSEQA, where NONE (84%) outperforms LONG (58%) by 26 percentage points—suggesting that for pattern-matching tasks, reasoning introduces noise rather than signal.

Why does verbose reasoning hurt OOD performance? We hypothesize three mechanisms. First, *error accumulation*: each reasoning step introduces a probability of error, and longer chains compound these errors [Wang et al., 2024]. Second, *domain-inappropriate reasoning*: when the model reasons at length about an unfamiliar domain, it may apply in-distribution heuristics that do not transfer, generating confident but incorrect intermediate conclusions. Third, *answer revision*: we observe that LONG frequently changes initially correct answers through excessive second-guessing, consistent with the “illusion of thinking” phenomenon [Shojaee et al., 2025].

The adaptive advantage and its limits. The ADAPTIVE policy achieves the best OOD-near accuracy (56%) and the smallest robustness gap (30 points) by dynamically allocating reasoning depth. However, its effectiveness is severely limited by poor confidence calibration (ECE = 0.258): the model reports high confidence even when incorrect, triggering the long-retry mechanism on only 4.4% of items. With better uncertainty estimation—e.g., probe-based methods using logit entropy [Ballon et al., 2026] or learned verifiers—we expect adaptive policies to show larger gains.

Confirmatory amplification. The confirmatory experiment on GPT-4.1-MINI reveals a striking amplification: OOD-near accuracy drops from 70% to 15% between SHORT and LONG. This suggests that stronger models, with greater capacity for generating plausible-sounding reasoning, may be *more* susceptible to overthinking. The overthinking problem may worsen, not improve, as models scale—an important consideration for deployment of frontier reasoning systems.

Practical implications. Our findings translate to direct deployment guidance:

- Default to short or medium reasoning for most tasks. Verbose reasoning (8+ steps) should be reserved for known in-distribution, high-difficulty problems.

- For OOD-robust deployment, shorter traces are safer. The $70\times$ efficiency advantage of concise reasoning makes this the dominant strategy for cost-sensitive applications.
- Adaptive policies with confidence-triggered escalation offer a practical middle ground, but require investment in calibration to realize their potential.
- Monitor robustness and calibration jointly. High confidence and long traces do not guarantee safer behavior—they may indicate the opposite.

Limitations. We acknowledge several limitations of this study. (1) *Sample size*: with $n = 50$ per dataset, confidence intervals are wide and some comparisons (e.g., MEDIUM vs. LONG) do not reach significance. Larger samples would sharpen estimates. (2) *Prompt-based control*: max-token limits and natural language instructions are imperfect length controls; actual trace length varies within each policy. (3) *Single model family*: results from the GPT-4.1 family may not generalize to other architectures such as Claude or Gemini. (4) *Deterministic decoding*: temperature = 0 eliminates sampling diversity; majority-voting approaches may show different patterns. (5) *Self-reported confidence*: the ADAPTIVE policy’s confidence mechanism is a behavioral signal, not a principled uncertainty estimate. Probe-based methods would provide more reliable uncertainty quantification. (6) *No RL-based control*: prompt-based length control is coarser than training-based approaches like L1/LCPO [Aggarwal and Welleck, 2025], which achieve finer-grained budget management.

Broader implications. Our results suggest that the relationship between reasoning depth and generalization is fundamentally different from the relationship between reasoning depth and in-distribution accuracy. This has implications for how we evaluate reasoning models: benchmark scores achieved with extensive reasoning may not predict real-world reliability under distribution shift. More broadly, the “more compute is better” paradigm for inference-time scaling needs to be qualified: additional compute helps only when it produces *relevant* reasoning, and domain shift can make extended reasoning counterproductive.

6 Conclusion

We present a controlled empirical study demonstrating that reasoning trace length has a non-monotonic relationship with both accuracy and OOD robustness. Across 1,250 evaluations on five benchmarks and two model sizes, we find that short reasoning (1–2 steps) consistently achieves the best or near-best OOD performance while using $9\times$ fewer tokens than verbose reasoning. The LONG policy catastrophically degrades OOD accuracy—by up to 55 percentage points on GPT-4.1-MINI—directly challenging the widespread assumption that more reasoning is uniformly beneficial.

The ADAPTIVE policy achieves the best OOD-near accuracy (56%) and smallest robustness gap (30 points), but its potential is limited by poor confidence calibration ($ECE = 0.258$). The clearest practical takeaway is direct: practitioners should avoid maximizing reasoning verbosity. For OOD-robust deployment, short-to-medium reasoning with adaptive fallback is optimal, and token-normalized performance strongly favors concise reasoning by up to $70\times$.

Future work should scale sample sizes for sharper statistical comparisons, extend to additional model families (Claude, Gemini) and task domains (code generation, multi-modal reasoning), and replace self-reported confidence with probe-based uncertainty estimates (logit entropy, learned verifiers) to unlock the full potential of adaptive trace-length control.

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